Chapter 5. Factor groups

5.3 The First Isomorphism Theorem

$$\begin{tikzcd} G \arrow[rr, "f"] \arrow[d, "\pi"'] & & H \\ G/\ker(f) \arrow[urr, dashed, "\tilde{f}"'] & & \end{tikzcd}$$

In this video, we will see a couple of examples.

Example 5.14. Let $G = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$. Show that $Z(G) \cong \mathbb{R}$ and $G/Z(G) \cong \mathbb{R} \times \mathbb{R}$.

We first prove that G is a group under matrix multiplication. In fact:

- G is closed under matrix multiplication:

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a + a' & b' + ac' + b \\ 0 & 1 & c + c' \\ 0 & 0 & 1 \end{pmatrix}$$

- The identity matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is in G, and
- G is closed under inverses: $\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -a & ac - b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix}$.

We compute now the centre of G:

If $\begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} \in Z(G)$, then

$$\begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix}, \quad \forall\, a, b, c \in \mathbb{R}.$$

$\Rightarrow \underline{b'} + ac' + b = b + \underline{a'c} + \underline{b'}$ for all $a, b, c \in \mathbb{R}$.

$\Rightarrow ac' = a'c$ for all $a, c \in \mathbb{R}$.

First take $c = 0$ and $a = 1$ to obtain that $c' = 0$. Secondly, take $c = 1$ and $a = 0$ to obtain $a' = 0$.

We have proved that

$$Z(G) = \left\{ \begin{pmatrix} 1 & 0 & z \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \;\middle|\; z \in \mathbb{R} \right\}$$

You can easily check that the map

$$\varphi : \mathbb{R} \longrightarrow Z(G)$$
$$r \longmapsto \begin{pmatrix} 1 & 0 & r \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

is a group isomorphism. It remains to show that $G/_{Z(G)} \cong \mathbb{R}\times\mathbb{R}$. To do this we are going to use the First Isomorphism Theorem. We need to find a group epimorphism $\varphi: G \longrightarrow \mathbb{R}\times\mathbb{R}$ with kernel $Z(G)$.

$$\varphi: G \longrightarrow \mathbb{R}\times\mathbb{R}$$

$$\varphi\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} = (a, c)$$

·) φ is surjective: for any $(r, s) \in \mathbb{R}\times\mathbb{R}$ we have $\varphi\begin{pmatrix} 1 & r & 0 \\ 0 & 1 & s \\ 0 & 0 & 1 \end{pmatrix} = (r, s)$.

•) φ is a group homomorphism:

product of matrices

$$\varphi\left(\begin{pmatrix}1 & a & b\\ 0 & 1 & c\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1 & a' & b'\\ 0 & 1 & c'\\ 0 & 0 & 1\end{pmatrix}\right) =$$

$$= \varphi\begin{pmatrix}1 & a+a' & b'+ac'+b\\ 0 & 1 & c+c'\\ 0 & 0 & 1\end{pmatrix} =$$

$$= (a+a', c+c') = (a, c) + (a', c') =$$

$$= \varphi\begin{pmatrix}1 & a & b\\ 0 & 1 & c\\ 0 & 0 & 1\end{pmatrix} + \varphi\begin{pmatrix}1 & a' & b'\\ 0 & 1 & c'\\ 0 & 0 & 1\end{pmatrix}$$

addition in $\mathbb{R}\times\mathbb{R}$

$\cdot)$ $\operatorname{Ker} \varphi \overset{?}{=} Z(G)$

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \in \operatorname{Ker} \varphi \Longleftrightarrow \varphi \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} = (0,0)$$

$$\Longleftrightarrow (a, c) = (0, 0) \Longleftrightarrow a = c = 0$$

$$\Longleftrightarrow \begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in Z(G)$$

By the 1st Iso Thm

$$G/Z(G) \cong \mathbb{R} \times \mathbb{R}.$$

Example 5.15. Suppose that G is a group of order 5, and $\varphi: \mathbb{Z}_{30} \to G$ is a group homomorphism.

Goal: Find $\operatorname{Ker}\varphi$

By the 1st Iso Thm

$$\mathbb{Z}_{30} / \operatorname{Ker}\varphi \cong \operatorname{Im}\varphi$$

By Lagrange's Theorem

$$|\operatorname{Im}\varphi| \,\Big|\, |G| = 5 \implies |\operatorname{Im}\varphi| = 1 \text{ or } 5$$

Thus:

$$\left| \mathbb{Z}_{30} / \operatorname{Ker} \varphi \right| = 1 \text{ or } 5.$$

•) If $\left| \mathbb{Z}_{30} / \operatorname{Ker} \varphi \right| = 1$, then

$$1 = \left| \mathbb{Z}_{30} / \operatorname{Ker} \varphi \right| = \frac{|\mathbb{Z}_{30}|}{|\operatorname{Ker} \varphi|}$$

$$\Rightarrow |\operatorname{Ker} \varphi| = |\mathbb{Z}_{30}| \Rightarrow \underline{\operatorname{Ker} \varphi = \mathbb{Z}_{30}}$$

·) If $\left| \mathbb{Z}_{30} / \mathrm{Ker}\,\varphi \right| = 5$, then

$$5 = \frac{|\mathbb{Z}_{30}|}{|\mathrm{Ker}\,\varphi|} = \frac{30}{|\mathrm{Ker}\,\varphi|} \Rightarrow |\mathrm{Ker}\,\varphi| = \frac{30}{5} = 6$$

By the Fundamental Thm of Finite Cyclic groups, there is only one subgroup of $\mathbb{Z}_{30}$ of order 6, which is $\mathrm{Ker}\,\varphi = 5\mathbb{Z}_{30}$.